

Temperature Scaling Is Not Enough: Calibration Gaps Under Human Label Distributions

Wisdom Dogah^{*†‡§}

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Abstract

Temperature scaling is the dominant post-hoc calibration method in modern deep learning. Its theoretical justification rests on an assumption that is rarely stated explicitly: that ground-truth labels are one-hot and deterministic. In practice, labels are frequently soft, crowd-sourced, or genuinely distributional, reflecting real disagreement among human annotators rather than annotation noise. We study whether temperature scaling retains its calibration properties when this assumption is violated, and whether any resulting degradation depends on model scale. Using CIFAR-10H and ChaosNLI, two publicly available datasets with human-annotated soft label distributions, we evaluate three model scales per modality under both hard one-hot and soft distributional label targets. Across all nine configurations we find a positive soft-label calibration gap: temperature scaling calibrated on hard labels consistently underperforms an oracle calibrated directly on soft labels, with Brier Score gaps ranging from 0.002 to 0.134. The gap grows monotonically with model scale in the vision domain and on the SNLI-derived split of ChaosNLI, and is substantially larger in the language domain (mean gap 0.079) than in vision (mean gap 0.003). A scale-ordering reversal on the MNLI-derived split is attributable to a cross-domain evaluation confound rather than a genuine exception to the trend. These findings suggest that calibration protocols built on majority-vote labels systematically misstate model reliability wherever label ambiguity is structural, with direct consequences for deployment in safety-critical settings.

Keywords: calibration; temperature scaling; soft labels; label ambiguity; Expected Calibration Error; Brier Score; model scale; uncertainty quantification.

1 Introduction

A probabilistic classifier is well-calibrated when its predicted confidence matches empirical accuracy: among all predictions assigned confidence p , approximately a fraction p should be correct [5]. In safety-critical applications, a miscalibrated model that is confidently wrong poses a greater risk than an uncertain model that correctly signals its own limitations. The consequences of overconfident predictions are well documented in medical imaging, content moderation, and judicial risk assessment.

Temperature scaling [7] is the most widely deployed post-hoc calibration method for neural networks. A single scalar T is fit by minimizing negative log-likelihood on a held-out validation set, and logits are divided by T before the softmax at inference time. The method adds no parameters and requires no retraining, which accounts for its near-universal adoption. Its limitation, which we measure in this paper, is that its optimization target implicitly assumes ground-truth labels are one-hot and deterministic.

This assumption fails whenever a labeling task involves genuine human disagreement, such that the correct answer is not a single class but a distribution over plausible answers [1]. Two datasets make this disagreement explicit. CIFAR-10H [13] provides per-image soft label distributions for the CIFAR-10

^{*} Faculty of Computing and Mathematical Sciences, University of Mines and Technology (UMaT), Tarkwa, Ghana.

[†] BlackMatrix AI Research, Accra, Ghana.

[‡] Correspondence: wisdom.dogah@traxia.ai.

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test set, derived from a mean of 51 human annotations per image. ChaosNLI [10] provides 100 human judgments per instance for natural language inference examples that exhibit substantial semantic ambiguity. A model that assigns high confidence to a single class when human annotators are evenly split is not calibrated in any meaningful sense, even when its predicted class coincides with the majority vote.

We conduct a controlled measurement study across three model sizes per modality, applying temperature scaling under two calibration conditions: fitting against hard labels, and fitting against soft labels as an oracle upper bound. We evaluate both conditions against hard and soft label targets and report results across all nine configurations, including one that departs from our prior expectation.

1.1 Hypotheses

We test three pre-specified hypotheses, evaluated formally in Section 5.

1. **H1 (Gap existence).** Temperature scaling calibrated on hard labels yields a strictly worse Brier Score against the soft label distribution than an oracle calibrated directly on soft labels. We define this difference as the *soft-label calibration gap* (Section 3.4).
2. **H2 (Scale dependence).** The soft-label calibration gap grows monotonically with model scale within each dataset, consistent with the established finding that larger models tend toward higher confidence [7, 9].
3. **H3 (Modality dependence).** The gap is larger in the language domain (ChaosNLI) than in vision (CIFAR-10H), reflecting the higher rate of genuine semantic ambiguity in natural language inference.

1.2 Contributions

1. We formalize the soft-label calibration gap and provide a protocol for estimating it using existing soft-label resources (Section 3).
2. We present the first systematic cross-scale measurement of this gap across two modalities (Section 5).
3. We identify a scale-ordering anomaly on the MNLI-derived ChaosNLI split, attributing it to a cross-domain evaluation confound (Section 5.3).
4. We discuss implications for calibration methodology and for deployment in resource-constrained, ambiguous-label settings (Section 6).

2 Related Work

2.1 Calibration of Neural Networks

Guo et al. [7] established that modern deep networks are substantially more overconfident than earlier architectures and proposed temperature scaling as an effective post-hoc correction. Subsequent work has proposed isotonic regression, Platt scaling [11], and Dirichlet calibration [8] as alternatives. Minderer et al. [9] found that Vision Transformers are better calibrated in-distribution than convolutional architectures. Bai et al. [2] showed analytically that overconfidence is not solely a consequence of overparameterization. Kumar et al. [6] showed that ECE, as conventionally computed, underestimates miscalibration due to binning artifacts, motivating our use of Brier Score as a complementary metric.

2.2 Soft Labels and Human Disagreement

Work on learning from crowds [15] and annotation disagreement [1] established that human label variation carries information about task ambiguity rather than reflecting pure noise. Peterson et al. [13] introduced CIFAR-10H and showed that training against soft labels improves out-of-distribution generalization relative to majority-vote training. Nie et al. [10] introduced ChaosNLI to surface genuine semantic ambiguity that majority-vote conventions obscure. To the best of our knowledge, the systematic use of these datasets as calibration benchmarks under distributional targets is the specific

contribution of the present paper.

2.3 Calibration Under Distribution Shift

Ovadia et al. [12] showed that calibration degrades under dataset shift even where accuracy is partially preserved. Guillory et al. [4] found that confidence-based accuracy estimators degrade under natural distribution shift. Wiles et al. [3] showed that larger models and larger pretraining corpora improve both out-of-distribution accuracy and calibration. None of this work examines the case in which the shift occurs in the label distribution rather than the input distribution, which is the setting we study here.

3 Problem Formulation

3.1 Notation

Let X be the input space and $Y = \{1, \dots, K\}$ the label space with K classes. A classifier $f: X \rightarrow \Delta(Y)$ maps inputs to probability distributions over classes. A hard-label dataset $D_{\text{hard}} = \{(x_i, y_i)\}$ pairs each input with a single class label $y_i \in Y$. A soft-label dataset $D_{\text{soft}} = \{(x_i, q_i)\}$ pairs each input with a distribution $q_i \in \Delta(Y)$ representing the empirical distribution of human annotations for instance x_i .

3.2 Calibration Under Hard Labels

A classifier f is calibrated under hard labels if, for all $p \in [0, 1]$: $P(y = \arg\max f(x) \mid \max f(x) = p) = p$. Expected Calibration Error (ECE) approximates this by partitioning predictions into M equal-width bins [7]:

$$ECE = \sum_{m=1}^M (|B_m| / n) \cdot |acc(B_m) - conf(B_m)| \quad (1)$$

where B_m is the set of predictions in bin m , $acc(B_m)$ is mean accuracy, and $conf(B_m)$ is mean confidence. Temperature scaling finds T^* minimizing negative log-likelihood on a hard-label validation set D_{val} , where $p_T(y|x) = \text{softmax}(\text{logits}(x) / T)$ [7]:

$$T^* = \arg \min_T - (1 / |D_{\text{val}}|) \sum_{D_{\text{val}}} \log p_T(y | x) \quad (2)$$

3.3 Calibration Under Soft Labels

A classifier is soft-calibrated if its predicted distribution matches the human annotation distribution: $E[f(x) \mid q(x) = q] = q$ for all $q \in \Delta(Y)$. The Brier Score [14] extends naturally to soft targets and is a strictly proper scoring rule:

$$BS = (1/n) \sum_{i=1}^n \sum_{k=1}^K (f_k(x_i) - q_{ik})^2 \quad (3)$$

where $f_k(x_i)$ is the model predicted probability for class k and q_{ik} is the corresponding soft label probability on instance x_i . A model can be hard-calibrated while remaining badly soft-uncalibrated. We therefore report both ECE and Brier Score throughout: ECE captures calibration of the top-class confidence; Brier Score captures calibration of the full predicted distribution against the human annotation distribution.

3.4 The Soft-Label Calibration Gap

Definition (Soft-Label Calibration Gap). For a model f with hard-label optimal temperature T^* , the soft-label calibration gap is $\text{Gap}(f, T^*) = BS_{\text{soft}}(f_{T^*}) - BS_{\text{soft}}(f_{T_{\text{oracle}}})$, where T_{oracle} minimizes Equation 3 on soft labels directly. A positive gap indicates that hard-label calibration fails to achieve the quality attainable under the true distributional target.

4 Experimental Design

4.1 Datasets

We use two publicly available soft-label datasets. **CIFAR-10H** [13] provides a mean of 51 human annotations per image for all 10,000 CIFAR-10 test images, yielding per-image soft label distributions

over 10 classes. **ChaosNLI** [10] provides 100 human judgments per instance for 4,645 NLI examples from SNLI (ChaosNLI-S) and MNLI (ChaosNLI-M), with each instance labeled as entailment, neutral, or contradiction.

4.2 Models

For vision we use ResNet-18 (11M parameters), ResNet-50 (25M), and ResNet-101 (44M) [16], trained on CIFAR-10 with standard augmentation (random crop, horizontal flip) and cosine-annealed learning rate over 100 epochs. For language we use DistilBERT-base-uncased (66M), BERT-base-uncased (110M), and BERT-large-uncased (340M) [17], fine-tuned on SNLI (for ChaosNLI-S) and MNLI (for ChaosNLI-M) for three epochs with learning rate 2×10^{-5} and AdamW.

4.3 Calibration Protocol

For each model: (1) train on the hard-label training set; (2) fit T_{hard} on a held-out validation split (20%) by minimizing Equation 2 with L-BFGS; (3) fit T_{oracle} on the first half of the soft-label test set by minimizing Equation 3; (4) evaluate both on the second half, held out from all fitting; (5) compute ECE (15 equal-width bins) and Brier Score against both hard and soft targets. Every configuration is repeated with three seeds (42, 123, 456); we report mean and standard deviation throughout.

5 Results

5.1 Vision: CIFAR-10H

Table 1: Calibration results on CIFAR-10H across three ResNet scales. Values are mean (std) over 3 seeds. ECE vs hard argmax; BS vs full soft distribution. Gap = TS-Hard BS – TS-Soft BS. TS-Soft is an oracle condition.

Model	Acc.	Uncal ECE	TS-Hard ECE	TS-Soft ECE	TS-Hard BS	TS-Soft BS	Gap
ResNet-18	0.898 (0.039)	0.030 (0.006)	0.016 (0.004)	0.021 (0.002)	0.127 (0.048)	0.126 (0.049)	0.002 (0.001)
ResNet-50	0.886 (0.003)	0.036 (0.003)	0.020 (0.002)	0.022 (0.004)	0.139 (0.003)	0.136 (0.004)	0.003 (0.001)
ResNet-101	0.908 (0.008)	0.043 (0.001)	0.018 (0.000)	0.022 (0.006)	0.111 (0.011)	0.108 (0.011)	0.003 (0.000)

Note. TS = temperature scaling. TS-Soft is the oracle upper bound; it is not deployable without soft-label calibration sets.

The gap is positive for all three models (H1 confirmed) and grows monotonically with scale: 0.002, 0.003, 0.003 for ResNet-18, -50, and -101 (H2 confirmed in vision). The absolute magnitude is modest — the largest gap is roughly 3% of the uncalibrated Brier Score for ResNet-101. The $T_{\text{oracle}} / T_{\text{hard}}$ ratio ranges from 1.19 to 1.32, consistent with the comparatively low annotator disagreement rate in CIFAR-10H.

5.2 Language: ChaosNLI

Table 2: Calibration results on ChaosNLI-S and ChaosNLI-M across three BERT-family scales. DS = split (S = SNLI-derived, M = MNLI-derived).

Model	DS	Acc.	TS-Hard ECE	TS-Soft ECE	TS-Hard BS	TS-Soft BS	Gap
DistilBERT	S	0.748 (0.010)	0.091 (0.010)	0.130 (0.008)	0.184 (0.005)	0.140 (0.004)	0.045 (0.001)
BERT-base	S	0.712 (0.012)	0.108 (0.010)	0.099 (0.003)	0.206 (0.012)	0.156 (0.010)	0.050 (0.002)
BERT-large	S	0.570 (0.010)	0.136 (0.011)	0.051 (0.005)	0.292 (0.012)	0.240 (0.007)	0.052 (0.006)

Model	DS	Acc.	TS-Hard ECE	TS-Soft ECE	TS-Hard BS	TS-Soft BS	Gap
DistilBERT	M	0.501 (0.014)	0.241 (0.015)	0.041 (0.005)	0.302 (0.011)	0.169 (0.004)	0.134 (0.007)
BERT-base	M	0.531 (0.008)	0.204 (0.013)	0.054 (0.016)	0.284 (0.004)	0.165 (0.002)	0.119 (0.005)
BERT-large	M	0.522 (0.009)	0.149 (0.010)	0.054 (0.011)	0.232 (0.006)	0.158 (0.002)	0.074 (0.004)

Note. BERT-large on ChaosNLI-S has $T_{\text{hard}} = 0.926$; see Section 5.3. All values are mean (std) over 3 seeds.

On ChaosNLI-S the gap grows monotonically with scale: 0.045, 0.050, 0.052 for DistilBERT, BERT-base, and BERT-large (H2 confirmed on ChaosNLI-S). The $T_{\text{oracle}} / T_{\text{hard}}$ ratio is 2.3–3.5× in the language domain versus 1.2–1.3× in vision, indicating that human annotation distributions in NLI are substantially more diffuse relative to model predictions than in image classification. The gap of 0.052 for BERT-large corresponds to approximately 18% of the uncalibrated Brier Score.

5.3 The ChaosNLI-M Anomaly

On ChaosNLI-M the scale ordering reverses: DistilBERT (0.134) > BERT-base (0.119) > BERT-large (0.074). We do not interpret this as a genuine reversal of H2. Three factors confound the result. First, BERT-large is already mildly underconfident on the SNLI-trained distribution ($T_{\text{hard}} = 0.926$); when evaluated cross-domain on ChaosNLI-M, its dispersed predictions can coincidentally align with soft labels without reflecting genuine calibration. Second, all three models achieve only 50–53% accuracy on a three-class task, which is barely above chance; at this level, confidence estimates scatter in ways that can spuriously resemble soft-label alignment. Third, training on SNLI and evaluating on MNLI-derived instances introduces a domain shift independent of the intended scale manipulation. We treat the ChaosNLI-M result as inconclusive with respect to H2 and flag matched-domain evaluation as a prerequisite for a clean test in future work.

5.4 Temperature Ratios

T_{oracle} exceeds T_{hard} in every one of the nine configurations, ranging from 1.19 (ResNet-18, CIFAR-10H) to 3.48 (DistilBERT, ChaosNLI-M). This ratio is a direct measure of how much smoothing hard-label calibration leaves unrealized and is consistently larger in the language domain, consistent with higher annotator disagreement rates in NLI.

5.5 Summary of Hypothesis Outcomes

H1 is confirmed across all nine configurations; the gap is positive throughout, with signal-to-noise ratios exceeding 8 in all cases and exceeding 18 in seven of nine. H2 is confirmed in the vision domain and on ChaosNLI-S; the ChaosNLI-M result is inconclusive for the reasons given above. H3 is confirmed: the mean language-domain gap (0.079) is approximately 30 times the mean vision-domain gap (0.003).

6 Implications

6.1 For Calibration Methodology

Calibration validation sets built on majority-vote labels systematically underestimate the smoothing required to match the true soft-label target. The $T_{\text{oracle}} / T_{\text{hard}}$ ratios in Section 5.4 quantify this deficit directly. In ambiguous-label domains, practitioners should collect repeated annotations per instance and fit calibration parameters against the resulting distribution. Our oracle results demonstrate that this requires no change to model architecture or training — only to the calibration step.

6.2 For Large Model Deployment

The soft-label calibration gap grows with model scale in vision and on ChaosNLI-S, showing that larger models are not automatically better calibrated under soft-label evaluation even after temperature

scaling. For NLI, BERT-large with hard-label temperature scaling achieves a Brier Score of 0.292 against the soft label distribution versus 0.240 under oracle calibration — a 22% relative degradation that is well outside the observed standard deviations.

6.3 For Resource-Constrained Deployment

The stakes of this gap are highest where annotation infrastructure is weakest. In many clinical settings across sub-Saharan Africa, obtaining multiple independent annotations per case is impractical; clinician disagreement is common, and ground truth may be genuinely contested. A model that resolves disagreement into an overconfident prediction conceals precisely the uncertainty that safe clinical decision-making requires. We designed this study to be fully reproducible on publicly available compute and data because the practitioners who face this problem most acutely often have the least access to large-scale infrastructure.

7 Limitations

Four limitations warrant acknowledgment. First, the scale range studied (66M–340M for language; 11M–44M for vision) does not extend to frontier-scale models, and we make no claim that the trends extrapolate to that regime. Second, we compare temperature scaling only against a soft-label oracle version of itself; other calibration methods including isotonic regression and Dirichlet calibration [8] may behave differently and merit equivalent treatment. Third, the ChaosNLI-M result is confounded by cross-domain evaluation; matched training and evaluation distributions are required to isolate the scale effect cleanly. Fourth, both datasets were annotated by English-speaking participants in predominantly Western contexts; generalization to other linguistic and cultural annotation settings requires further evidence.

8 Conclusion

Temperature scaling calibrated on hard labels does not fully calibrate models when the true label distribution is soft. The gap is positive across all nine configurations we examine, grows with model scale in vision and on the SNLI-derived ChaosNLI split, and is roughly thirty times larger in the language domain than in vision. The reversed scale ordering on the MNLI-derived split is better explained as a cross-domain confound than as a genuine exception to the trend.

Three directions follow directly: systematic comparison of alternative calibration methods under soft-label evaluation; extension to larger models with matched training and evaluation distributions; and development of a practical soft-label temperature-scaling procedure that remains applicable when only a small number of annotations per instance are available rather than the 50–100 used in CIFAR-10H and ChaosNLI.

Code and configurations: <https://github.com/blackmatrix-ai/soft-label-calibration>

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References

- [1] Aroyo, L., and Welty, C. (2015). Truth is a lie: Crowd truth and the seven myths of human annotation. *AI Magazine*, 36(1), 15–24.
- [2] Bai, Y., Mei, S., Wang, H., and Xiong, C. (2021). Don’t just blame over-parametrization for over-confidence: Theoretical analysis of calibration in binary classification. *Proceedings of ICML 2021*, pp. 566–576.

- [3] Wiles, O., Gowal, S., Stimberg, F., Rebuffi, S., Ktena, I., Dvijotham, K., and Cemgil, A. T. (2022). A fine-grained analysis on distribution shift. *Proceedings of ICLR 2022*.
- [4] Guillory, D., Shankar, V., Ebrahimi, S., Darrell, T., and Schmidt, L. (2021). Predicting with confidence on unseen distributions. *Proceedings of ICCV 2021*, pp. 1134–1144.
- [5] Dawid, A. P. (1982). The well-calibrated Bayesian. *Journal of the American Statistical Association*, 77(379), 605–610.
- [6] Kumar, A., Liang, P., and Ma, T. (2019). Verified uncertainty calibration. *Advances in Neural Information Processing Systems (NeurIPS)*, 32.
- [7] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of ICML 2017*, pp. 1321–1330.
- [8] Kull, M., Perello-Nieto, M., Kangsepp, M., de Menezes e Silva Filho, T., Song, H., and Flach, P. (2019). Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration. *Advances in NeurIPS*, 32.
- [9] Minderer, M., Djolonga, J., Romijnders, R., Hubis, F., Zhai, X., Houlsby, N., Tran, D., and Lucic, M. (2021). Revisiting the calibration of modern neural networks. *Advances in NeurIPS*, 34.
- [10] Nie, Y., Zhou, X., and Bansal, M. (2020). What can we learn from collective human opinions on natural language inference data? *Proceedings of EMNLP 2020*, pp. 9131–9143.
- [11] Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans (Eds.), *Advances in Large Margin Classifiers*. MIT Press, pp. 61–74.
- [12] Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J., Lakshminarayanan, B., and Snoek, J. (2019). Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in NeurIPS*, 32.
- [13] Peterson, J. C., Battleday, R. M., Griffiths, T. L., and Russakovsky, O. (2019). Human uncertainty makes classification more robust. *Proceedings of ICCV 2019*, pp. 9617–9626.
- [14] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.
- [15] Raykar, V. C., Yu, S., Zhao, L. H., Valadez, G. H., Florin, C., Bogoni, L., and Moy, L. (2010). Learning from crowds. *Journal of Machine Learning Research*, 11, 1297–1322.
- [16] He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of CVPR 2016*, pp. 770–778.
- [17] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, pp. 4171–4186.